

Arnob Alam
ArnobLAlam@gmail.com
ArnobLAlam.com

Education

- 2025 **Ph.D.**, Economics, American University
 Title: Causal Impacts of Financial Market Regulation: Evidence From Central Clearing and Methodological Advances in Difference-In-Differences
 Advisors: Amos Golan (chair), Robin Lumsdaine, Kara Reynolds, Michael Stutzer
- 2016 **M.A.**, Economics, American University
- 2010 **B.Sc.**, Microelectronics Engineering & Economics, Rochester Institute of Technology

Job Market Paper

Alam, Arnob. *The Impact of Mandatory Central Clearing on Interest Rate Swaps Pricing*, 2025.

Abstract: Mandatory central clearing was the keystone derivatives market reform in the Dodd-Frank era. I examine the impact of the mandate on interest rate swaps pricing. Using data from Depository Trust & Clearing Corporation's Swaps Data Repository around the time of the central clearing mandate implementation and using Canadian-dollar denominated contracts, which were not subject to the clearing mandate as a comparison group, a 13-14 bps increase in swap premia is observed for U.S.-dollar-denominated contracts. A similar increase of 7-8 bps is observed for contracts denominated in British Pounds (subject to the clearing mandate) compared to those denominated in Swiss Francs (not subject to the clearing mandate). The results agree with theoretical models that show swaps become more valuable as counterparty risk is reduced, and existing literature that shows mandatory clearing had similar effects on equities markets. The results are robust to additional controls for contract characteristics and alternative functional specifications.

Publications

- Arslan, Shirin, **Alam, Arnob**, Maria S Floro, Seung-Eun Cha, and Eunhye Kang. "Care workers' sense of responsibility, working conditions, and the quality of care in South Korea". *Philippine Review of Economics* (Online ISSN 2984-8156) 60, no. 1 (2023): 191–222.
- Cha, Seung-Eun, Eunhye Kang, Maria Floro, Shirin Arslan, and **Arnob Alam**. "When caregiving becomes burden: the gap between preferred amount of care time and actual amount of care time". *Social Indicators Research* 170, no. 2 (2023): 375–397.
- Henderson, Heath, and **Alam, Arnob**. "The structure of risk-sharing networks". *Empirical Economics* 62, no. 2 (2022): 853–886.

Presentations

- Quality of Care, Commitment Level, and Working Conditions: Understanding the Care Workers' Perspective.* 30th International Association for Femenist Economists Annual Conference (Geneva, Switzerland), June 2022.

The Structure of Risk-Sharing Networks. 2nd International Conference on Complexity And Policy Studies (Arlington, VA, USA), Apr. 2018.

Working papers

Everson, Kelly R., and **Alam, Arnob**. *NEWS for Riyadh, Saudi Arabia: Population-based Walking, Physical Activity, and Neighborhood Characteristics*, 2025.

Professional Experience

- 2021–present **Operations Research Analyst**, *Department of Defense*, Arlington, VA
Developed readiness optimization and force-structure models for the Army National Guard; advised senior leadership on stationing, divestment, and resource-allocation decisions.
- 2024–2025 **Short Term Consultant**, *World Bank*, Washington, DC
Designed sampling weights for the Riyadh Walkability Survey using a stratified design; established baseline physical-activity and health indicators for follow-up surveys.
- 2018–2020 **Senior Analyst**, *Right Sized Solutions Inc.*, Arlington, VA
Built models of personnel retention, recruitment, and promotion for the Army National Guard; advised on the impacts of policy, marketing, and incentives; allocated resources across 54 states and territories to meet strength and readiness goals.
- 2018–2018 **Research Consultant (short term)**, *Talascend Inc.*, Herndon, VA
Validated data quality for a new education-data product developed by the National Student Clearinghouse.
- 2015–2016 **Data Scientist**, *WeddingWire Inc.*, Chevy Chase, MD
Performed A/B testing and machine-learning analyses of marketing and pricing strategies; estimated median wedding spend by vendor.
- 2014–2015 **Senior Data Engineer**, *GovTribe Inc.*, Roslyn, VA
Developed large-scale web-scraping and text-analysis systems for federal procurement data; built and maintained NoSQL databases (MongoDB, Elasticsearch) for data storage and retrieval.

Research Experience

- 2019–2022 **Researcher**, *Care Work and the Economy Project*
Analyzed demand for eldercare and childcare, and investigated determinants of care quality in South Korea using information-theoretic (info-metrics) methods.
- 2017 **Research Intern**, *Division of Enforcement, Commodity Futures Trading Commission (CFTC)*
Monitored new commodity exchanges to ensure market-making incentive schemes did not encourage wash trading, spoofing, or other disorderly market activities.

- 2016–2018 **Statistics Consultant**, *Center for Teaching, Research and Learning*, American Univ.
Advised faculty and students across disciplines on appropriate statistical methods, research design, and data analysis.
- 2016 **Research Intern**, *Office of the Chief Economist (OCE)*, CFTC
Examined the industrial organization of the non-deliverable forwards (NDF) market using a proprietary regulatory dataset.
- 2016 **Research Assistant**, *Drake University* (Advisor: *Dr. Heath Henderson*)
Investigated the structure of real-world risk-sharing networks in developing-country settings and compared empirical patterns to theoretical network models.
- 2015 **Research Intern**, *Office of the Chief Economist (OCE)*, CFTC
Evaluated market quality in the NDF market, focusing on pricing, liquidity, volatility, and use-case analysis.

Teaching

2019–2021	Adjunct Lecturer, Economics for Business (ECON-150)	American University
2018–2019	Graduate Teaching Assistant, Financial Management (FIN-614)	American University

Tools & Software

- **Languages:** Python, R, Julia, GAMS, Haskell, Clojure, SAS, Stata, Perl, SQL, Bash/Shell scripting, YAML, JSON, XML
- **Tools:** Google BigQuery, Databricks, Apache Spark, Hadoop, SQL Server, PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, Power BI, Tableau, QLIK, igraph, networkx, DataTable, Pandas, Scipy, Numpy, ggplot, matplotlib
- **Techniques:** Bayesian Methods, Tree-based Methods (Classification and Regression Trees, Random Forests, Boosting and Bagging), Regularization (LASSO, Ridge, Elasticnet), Clustering (Hierarchical Clustering, K-Nearest Neighbors) Smoothing Methods (LOESS, Splines), Natural Language Processing (Named Entity Recognition, Sentiment Analysis), Unsupervised Learning (Principal Components, Factor Analysis, Correspondence Analysis)

Last updated: October 13, 2025